Internship Report

on

Software Engineering Internship at Oracle Oracle Communications Global Industry Unit (CGIU) Oracle Communications Session Border Controller (SBC)

by

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Submitted to

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3 messages

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Hi Team,

We would like to inform you that the below students are currently undertaking a 6-month internship with Oracle, which commenced on **Jan 7, 2026** and is scheduled to conclude on **July 10, 2026.**

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Thanks,

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Declaration

I hereby declare that this internship report titled *"Software Engineering Internship at Oracle – Oracle Communications Global Industry Unit (CGIU) – Oracle Communications Session Border Controller (SBC)"* is an authentic account of the training, learning, analysis, and work carried out by me during my internship at Oracle. The content is based on my own experience and work performed under the guidance of mentors and team members during the internship period.

No part of this report has been submitted elsewhere for any academic purpose. Internal product-specific information has been described only at a high level and in a confidentiality-preserving manner. The report focuses on academic learning outcomes, engineering exposure, tools used, responsibilities handled, and the professional contribution made during the internship.

Signature of the Student

Jay Gupta (12212116)

Date: May 2026

Acknowledgement

I would like to express my sincere gratitude to **Oracle** for providing me with the opportunity to intern and contribute to work associated with a widely used enterprise communication product environment. I am especially thankful to the Oracle Communications Global Industry Unit (CGIU), the Session Border Controller team, my mentors, team leads, and colleagues for their continuous guidance, patience, and support in helping me grow both technically and professionally.

The Knowledge Transfer (KT) sessions, structured training modules, product walkthroughs, code-review discussions, and day-to-day mentorship were instrumental in helping me understand professional software engineering practices in a real-world, production-grade environment. The experience strengthened my appreciation for technical depth, confidentiality, accountability, evidence-based analysis, and collaborative problem solving.

I also extend my gratitude to the faculty and evaluation committee at the National Institute of Technology Kurukshetra for their academic support and for structuring this internship as a formal learning opportunity.

Jay Gupta

Executive Summary

This report presents a detailed account of the Software Engineering Internship undertaken by **Jay Gupta** at **Oracle**, in association with the Oracle Communications Session Border Controller (SBC) team within the Oracle Communications CGIU context. The internship spanned **7 January 2026** to **10 July 2026** and provided structured exposure to enterprise software engineering, telecom product validation, Oracle Linux migration testing, and professional engineering practices.

The internship included onboarding, foundational technical training, product-focused SBC exposure, and team-based validation work. The main technical focus was on **SBC validation using the *Anvil* Framework**, **Oracle Linux migration testing (OL7, OL8, and OL9)**, and ***TesView*-based monitoring and result analysis**. A major part of Phase 4 also covered **Diameter-based billing testing** and understanding high-level call flows used in validation.

Keywords: Oracle Communications, CGIU, Session Border Controller, *Anvil* Framework, TesView, SBC Validation, OL7–OL9 Migration, Diameter Testing, Oracle Linux, Regression Testing.

Contents

Internship Confirmation Certificate **1**

Declaration **3**

Acknowledgement **4**

Executive Summary **5**

1 Introduction **10**

1.1 Background 10

1.2 Motivation 10

1.3 Objectives of the Internship 10

1.4 Scope and Confidentiality 11

1.5 Report Organisation 11

2 Organisation Profile: Oracle and Oracle CGIU **12**

2.1 Oracle Corporation Overview 12

2.2 Oracle Communications Global Industry Unit (CGIU) 12

2.3 Oracle Communications Session Border Controller 13

2.4 Engineering Culture and Professional Exposure 13

3 Internship Timeline and Training **14**

3.1 Phase 1: Onboarding, Laptop Setup, and Compliance Training 14

3.2 Phase 2: Foundational Technical Training 15

3.3 Phase 3: Product-Focused SBC Training 15

3.4 Phase 4: Team-Based Work . . . 16
3.4.1 Anvil Framework and Suite Execution . . . 16
3.4.2 Observed Validation Suite Structure . . . 16
3.4.3 Diameter-Based Billing Validation . . . 17
3.4.4 Observed Call Flow During Validation . . . 17
3.4.5 Oracle Linux Migration Testing (OL7 to OL9) . . . 18
3.4.6 TesView-Based Monitoring and Analysis . . . 18

4 Technical Background **20**
4.1 Session Border Controller (SBC) . . . 20
4.2 Anvil Framework . . . 20
4.3 Diameter-Based Billing Validation . . . 21
4.4 TesView Monitoring . . . 21
4.5 Oracle Linux Migration Testing . . . 22

5 Tools and Technologies Used **23**

6 Learning Outcomes and Skills Gained **24**
6.1 Technical Skills . . . 24
6.2 Analytical and Problem-Solving Skills . . . 24
6.3 Professional Skills . . . 25

7 Conclusion **26**
7.1 Future Scope . . . 26

References **27**

A Confidentiality Notes **28**

Appendix A: Confidentiality Notes **28**

List of Tables

2.1 Oracle CGIU / Oracle Communications Focus Areas and Internship Relevance 13

3.1 Internship Phase Summary 14

3.2 Training Modules and Relevance to Internship Work 15

5.1 Tools, Technologies, and Learning Relevance 23

List of Figures

3.1 High-level SBC and Diameter billing validation call flow 18

Chapter 1

Introduction

1.1 Background

Modern communication networks require software systems that process large volumes of signalling and media-related traffic reliably, securely, and with low latency. Enterprises and communication service providers depend on such systems to support voice, video, messaging, unified communications, mobile services, and cloud-connected communication platforms.

Oracle is a global technology company known for database systems, cloud infrastructure, enterprise applications, Java, and industry-specific solutions. Within Oracle, **Oracle Communications** focuses on products and solutions for communication service providers and enterprises, including signalling, routing, network security, enterprise communications, and cloud-native transformation.

1.2 Motivation

The motivation behind this internship was to gain exposure to real-world software validation and systems engineering beyond academic coursework. Concepts such as operating systems, Linux environments, networking, test automation, and version management become much more meaningful when applied to a large production-grade telecom product.

1.3 Objectives of the Internship

1. Understand Oracle workplace practices, compliance requirements, confidentiality expectations, and enterprise engineering culture.

2. Strengthen core technical skills through structured training in UNIX/Linux, networking, Git, DevOps, and observability.

3. Gain a product-level understanding of Oracle Communications Session Border Controller and its role in communication networks.

4. Execute and analyse *Anvil* Framework validation suites across Oracle Linux environments.

5. Compare suite execution behaviour across OL7, OL8, and OL9 and document compatibility impacts.

6. Use *TesView*-based monitoring to track validation results and support rerun management.

7. Study Diameter-based billing flows at a high level for validation and troubleshooting.

1.4 Scope and Confidentiality

The scope of this report is limited to academic documentation of the internship experience. Product-specific architectural details, source-code fragments, exact suite output logs, internal workload identifiers, customer data, and confidential Oracle information have not been disclosed.

1.5 Report Organisation

The remaining sections describe the Oracle and Oracle CGIU profile, the internship timeline, technical background, work undertaken, results, tools and technologies, learning outcomes, challenges, conclusion, future scope, references, and confidentiality notes.

Chapter 2

Organisation Profile: Oracle and Oracle CGIU

2.1 Oracle Corporation Overview

Oracle Corporation is a multinational enterprise technology company providing cloud infrastructure, database technologies, enterprise applications, developer tools, analytics, and industry solutions.

2.2 Oracle Communications Global Industry Unit (CGIU)

Oracle Communications Global Industry Unit (CGIU) represents the communications-focused industry engineering context of the internship. Oracle Communications serves communication service providers and enterprises with solutions related to network security, enterprise communications, signalling, orchestration, assurance, and cloud-native communication services.

Table 2.1: Oracle CGIU / Oracle Communications Focus Areas and Internship Relevance

Focus Area	High-Level Description	Relevance to Internship Learning
Signalling and routing	Technologies that manage communication control paths.	Provided context for SBC behaviour and validation.
Enterprise communications	Solutions that support collaboration and protected voice connectivity.	Made the role of session border control easier to understand.
Unified operations	Operational tools for observability and automation.	Connected training in Grafana, Prometheus, and TesView to real product operations.
Cloud-native services	Modern telecom services and network functions.	Helped relate software engineering to large-scale telecom systems.

2.3 Oracle Communications Session Border Controller

Oracle Communications Session Border Controller is an industry-leading product for fixed line, mobile, and over-the-top services, operating on Oracle purpose-built hardware platforms or general-purpose servers.

2.4 Engineering Culture and Professional Exposure

The Oracle engineering environment demonstrated how professional teams approach complex product work. Technical discussions were evidence-driven, and even incremental changes required careful reasoning.

Chapter 3

Internship Timeline and Training

The internship progressed through four major phases: onboarding and compliance, foundational technical training, product-focused SBC training, and team-based engineering work.

Table 3.1: Internship Phase Summary

Phase	**Period**	**Focus Area**	**Outcome**
1	7 Jan – 16 Jan 2026	Onboarding, laptop setup, access, compliance training	Prepared for secure and responsible work in an enterprise environment
2	19 Jan – 13 Mar 2026	Foundational technical training	Strengthened programming, OS, networking, cloud, and observability foundations
3	Mid-Mar 2026	Product-focused SBC training	Built high-level understanding of SBC architecture, *Anvil*, TesView, and OL environments
4	1 Apr – 10 Jul 2026	Team-based validation, migration testing, and billing-flow analysis	Executed *Anvil* suites, performed OL7–OL9 migration analysis, and monitored results via TesView

3.1 Phase 1: Onboarding, Laptop Setup, and Compliance Training

- Completed internship onboarding formalities and became familiar with Oracle internal systems and communication practices.

- Set up the company laptop and configured the basic software environment required for daily work.
- Completed mandatory compliance, security, and workplace training modules.
- Understood expectations related to information security, data confidentiality, responsible use of internal resources, and professional workplace conduct.

3.2 Phase 2: Foundational Technical Training

Table 3.2: Training Modules and Relevance to Internship Work

Training Module	**Period**	**Technical Relevance**
UNIX and Linux Essentials	2–4 Feb	Command-line work, environment navigation, and log inspection.
Networking Basics	5–6 Feb	Context for communication products and protocol-oriented thinking.
Git	9 Feb	Version control, collaboration, and code review.
C++ and Data Structures	10–12 Feb	Class hierarchies, object lifetime, and code structure understanding.
Debugging using GDB-style tools	13, 16 Feb	Runtime investigation, stack inspection, and issue diagnosis.
Multithreading and IPC	17–19 Feb	Concurrent execution, worker threads, message passing, and communication between components.
OCI DevOps	23–27 Feb	CI/CD, build pipelines, deployments, and OCI software delivery.
Monitoring and Alerting	9–13 Mar	Observability concepts using Prometheus and Grafana.

3.3 Phase 3: Product-Focused SBC Training

- Attended SBC-focused product sessions to understand the purpose and high-level architecture of Oracle Communications Session Border Controller.
- Studied *Anvil* Framework structure, suite organisation, configuration files, and execution workflows.
- Understood *TesView* monitoring, result visualisation, rerun management, and validation summary interpretation.

- Studied Oracle Linux environment differences (OL7, OL8, OL9) and how platform generation affects product validation.

3.4 Phase 4: Team-Based Work

The fourth phase involved practical contributions to the SBC validation team. The work mainly focused on *Anvil* Framework suite execution, Diameter-based billing validation, Oracle Linux migration testing (OL7, OL8, and OL9), and *TesView*-based result monitoring.

3.4.1 Anvil Framework and Suite Execution

A major part of the work involved understanding the *Anvil* Framework structure and execution workflow used for SBC validation and Diameter billing scenarios. The framework organizes reusable validation suites, setup files, execution scripts, logs, and rerun data in a structured format.

The work included:

- Reviewing suite folders, setup scripts, and configuration files.
- Understanding how validation scenarios are executed and monitored.
- Observing rerun workflows and failure analysis using logs and summaries.
- Comparing execution behavior across Oracle Linux 7, 8, and 9 environments.

3.4.2 Observed Validation Suite Structure

The following structure was commonly observed during validation and billing-related testing:

```
automation/
 scripts/
     ACCOUNTING/
         DIAMETER/SIP/
             DIAMETER_MANIPULATION_RULES_FOR_RF_INTERFACE/
                 logs/
                 scenario/
                 *.tcl
```

```
        *.config
        *.setup

    DIAMETER_RF_NEW_AVPS/
    DIAMETER_RF_EVENT_ACR/
    DIAMETER_RF_IOI_AVP/
    DIAMETER_NUMBER_PORTABILITY/
    other validation suites
```

The structure helped in understanding how setup, execution, logging, and rerun workflows are separated inside the framework.

3.4.3 Diameter-Based Billing Validation

A significant part of the validation work was related to Diameter-based billing testing. The testing focused on validating request-response handling, accounting flows, Diameter AVPs, and SBC billing-related communication scenarios.

Observed validation areas included:

- Diameter manipulation rules for RF interfaces.
- Accounting request and response validation.
- AVP-related validation scenarios.
- Diameter event ACR flows.
- Number portability and routing-related Diameter scenarios.
- Billing-related SBC validation across multiple Linux environments.

3.4.4 Observed Call Flow During Validation

The validation logs contained SIP and Diameter call-flow sequences used for billing verification and accounting validation. The observed flow involved SIP INVITE processing, provisional responses, ACK handling, Diameter accounting flow processing, and teardown validation.

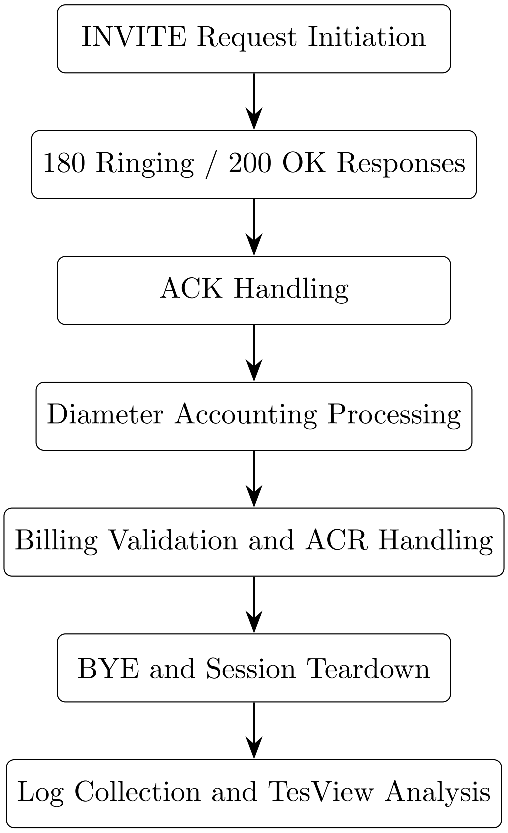

Figure 3.1: High-level SBC and Diameter billing validation call flow

3.4.5 Oracle Linux Migration Testing (OL7 to OL9)

Another important responsibility involved comparing validation-suite behavior across Oracle Linux 7, Oracle Linux 8, and Oracle Linux 9 environments.

The work included:

- Running and comparing the same validation suites across OL7, OL8, and OL9.
- Observing package-level and dependency-level differences.
- Identifying failures caused by environment or platform differences.
- Reviewing service startup behavior and runtime differences across Linux versions.
- Supporting migration validation and rerun analysis.

3.4.6 TesView-Based Monitoring and Analysis

TesView was used to monitor suite execution history, rerun behavior, and recurring validation failures.

The work involved:

- Reviewing validation summaries and rerun history.
- Comparing logs across multiple execution runs.
- Identifying recurring billing and Diameter-related failures.
- Monitoring validation status across Oracle Linux environments.
- Supporting debugging and rerun planning discussions.

Chapter 4

Technical Background

4.1 Session Border Controller (SBC)

Oracle Communications Session Border Controller (SBC) is a telecom network element used to manage signaling and media communication at network boundaries. The SBC helps provide secure and reliable communication between different networks by handling SIP signaling, session routing, interoperability, security enforcement, and media-related processing.

In enterprise and telecom environments, the SBC operates in performance-sensitive and high-availability conditions where validation and stability are important. During the internship, exposure to SBC workflows helped in understanding how signaling flows, accounting scenarios, and validation suites are connected to real telecom communication systems.

4.2 Anvil Framework

The *Anvil* Framework is an internal automation and validation framework used for executing SBC-related validation suites across different environments. The framework organizes suite execution, setup configuration, logging, rerun handling, and validation monitoring in a structured workflow.

The framework commonly uses:

- Setup and prerequisite scripts.
- Environment-specific configuration files.
- Validation and accounting test suites.

- Structured logging and rerun directories.
- Automated execution and result collection workflows.

During the internship, the framework was used for running Diameter and SBC validation scenarios across Oracle Linux environments and for reviewing execution behavior through logs and monitoring tools.

4.3 Diameter-Based Billing Validation

Diameter is a telecom signaling protocol used for authentication, authorization, and accounting workflows. In SBC validation environments, Diameter-based testing is commonly associated with billing verification, accounting request handling, AVP validation, and communication-flow correctness.

The internship work involved reviewing Diameter-related validation scenarios at a high level, including:

- Diameter accounting request and response handling.
- Diameter RF interface validation scenarios.
- AVP-related validation and manipulation rules.
- Accounting flow verification during SIP session handling.
- Billing-related validation behavior during call setup and teardown.

Understanding these flows helped in interpreting validation logs and identifying recurring behavior during rerun analysis.

4.4 TesView Monitoring

TesView is used for reviewing validation history, rerun summaries, execution logs, and recurring failures generated during *Anvil* Framework execution. It helps organize execution results and provides visibility into validation behavior across different environments and reruns.

The monitoring workflow mainly involved:

- Reviewing pass/fail execution summaries.

- Comparing rerun outputs across multiple executions.
- Tracking recurring failures during billing and Diameter validation.
- Observing environment-specific validation differences across Oracle Linux versions.

4.5 Oracle Linux Migration Testing

Oracle Linux migration testing involved comparing validation-suite behavior across Oracle Linux 7, Oracle Linux 8, and Oracle Linux 9 environments. Since platform generations introduce package, dependency, service, and runtime differences, migration validation is important for maintaining stable product behavior.

The migration-related work mainly focused on:

- Running the same validation suites across OL7, OL8, and OL9.
- Identifying package and dependency-level differences.
- Observing service startup and runtime behavior changes.
- Comparing validation logs and rerun behavior across environments.
- Supporting migration-related validation and compatibility analysis.

Chapter 5

Tools and Technologies Used

Table 5.1: Tools, Technologies, and Learning Relevance

Category	**Technology / Tool**	**Usage During Internship**
Test Automation	*Anvil* Framework	Suite setup, execution, rerun handling, and validation across OL environments.
Monitoring	TesView	Reviewing suite history and tracking rerun results.
OS Platforms	Oracle Linux 7/8/9	Comparative validation and migration compatibility analysis.
Systems	UNIX/Linux	Command-line work, environment navigation, and log inspection.
Version Control	Git	Collaborative development and change management.
Cloud/DevOps	OCI DevOps	CI/CD, build pipelines, deployments, and OCI software delivery.
Observability	Grafana + Prometheus	Monitoring, metrics dashboards, and alerting.
Product Domain	Oracle Comms SBC	Telecom product understanding and validation scenario interpretation.

Chapter 6

Learning Outcomes and Skills Gained

6.1 Technical Skills

- Hands-on experience with *Anvil* Framework.
- Practical understanding of validation result analysis.
- *TesView* monitoring.
- Oracle Linux platform awareness.
- Migration testing methodology.
- UNIX/Linux proficiency.
- Networking fundamentals relevant to SBC communication behaviour.
- High-level understanding of Diameter-based billing validation.

6.2 Analytical and Problem-Solving Skills

- Ability to distinguish environment-induced failures from product defects.
- Ability to compare results across multiple platform environments.
- Understanding of how platform generation changes affect product validation.
- Improved root-cause analysis.
- Ability to document technical findings clearly.

6.3 Professional Skills

- Awareness of Oracle onboarding, compliance, confidentiality, and secure handling of internal product information.
- Improved technical communication through KT sessions and structured documentation.
- Understanding of team collaboration, mentor interaction, and review-oriented thinking.

Chapter 7

Conclusion

The internship at **Oracle** was a valuable learning experience that combined structured training, product understanding, and practical engineering contribution. The onboarding phase introduced Oracle professional standards and compliance expectations. The foundational training phase strengthened skills in Linux, networking, debugging, DevOps, and observability. The product-focused SBC training connected these concepts to a real Oracle Communications product operating in performance-sensitive communication environments.

The team-based engineering phase provided hands-on exposure to enterprise validation workflows. Executing *Anvil* Framework suites across Oracle Linux environments, analysing results systematically, performing cross-environment migration comparisons, monitoring validation history through *TesView*, and reviewing Diameter-based billing flow testing demonstrated how software quality is maintained across platform generations in a large telecom product.

7.1 Future Scope

- Expanded OL9 validation coverage.
- Automated migration regression detection.
- OL9 certification documentation.
- TesView trend analysis.
- Suite execution optimisation.
- Knowledge transfer documentation.

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Appendix A

Confidentiality Notes

- Exact *Anvil* suite logs, raw test output, pass/fail counts, benchmark values, workload identifiers, internal screenshots, and source-code fragments are not included.
- Oracle product behaviour and architecture are discussed only at a high level.
- The internship confirmation certificate is included as supporting academic documentation.
- Any additional internal diagram or numerical result should be added only after explicit approval.